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Dirty Spokes Dirtbike Rental Management System

Feedback by TAs and Peer Reviewers

Review by: Michael Santora

Hi Spencer and Matthew!

This is really impressive work on the overview. You painted an excellent picture of the business model while managing to highlight the key operational issues a DBMS could help solve. The objective in paragraph three is clearly stated: creating unification while ensuring data accuracy and real-time tracking. Finally, the included image was also a nice touch to make the scenario feel more "real!"

The four entities you've identified are logical, and it was a smart design choice to make each table's ID field specific (e.g., customerID and rentalID) rather than using a generic "ID" across the board. To further refine your schema, I suggest the following:

- For data types like VARCHAR or INT, consider specifying the maximum length or display width.
- It may be worth noting which INT attributes should be signed vs. unsigned.
- Perhaps another table to house Employee would be beneficial for this DBMS.
- I see you've noted the Rentals table as an intersection table to facilitate the M:N relationship between Bikes and Customers. However, since it houses specific event data, it also could function as a transaction table. It may be worth pivoting to make Rentals a standard transaction table and create a dedicated intersection table (e.g., CustomersBikes) containing only the Foreign Keys. Regardless, I believe your current setup to be quite functional for this.

I really like your ERD model because it shows perfect consistency with your written overview. You correctly followed the standard naming conventions by keeping entities plural and attributes singular (e.g., Customers, customerID). The tables and respective attributes match the outline perfectly, and the Crow's Foot notation makes the logic very easy to follow!

Lastly, thank you for clarifying the sources you leveraged for this stage of the project.

Nice work, both!

No external sources were leveraged in my review at this time.

Review by: Joshua York

I agree, Dirty Spokes, dirt bike rentals could use a database and web site to track the rentals, customers, and bikes from the overview. 50 bikes over 2500 annual rentals could be difficult to manage with paper receipts. Yes, the overview lists specific facts that support the use of the DB and website. Specifically, the overview goes over lost records or duplicate records, a lack of accurately tracking rental hours, and which customer had which bike using their current method.

There are 4 entities described and each represent a single idea. Bikes, Brands, Customers, and Rentals are all separate ideas. The Rentals table represents the M:N relationship between Customers and Bikes but it adds additional attributes that make it an entity. The relationship between the entities is described well and there are 1:M and M:N relationships present.

The outline accurately details the purpose and attributes (including datatypes and constraints). The relationships are present and described. Bikes to Brands is described as a Brand can have many Bikes. The relationship between Bikes to Brands I believe should be mandatory instead of optional. As a bike must have a brand.

In the ERD, there's a connection from Customer to Bikes (M:N) that should be removed. It's not referenced in the database outline, and that relationship is covered by the Rentals table. Cardinality is missing in the same way as the outline describes the relationships. I'm not certain, but these look like mandatory lines instead of optional lines.

There is a consistent naming convention. The table names are plural as required (Bikes, Customers, Rentals, Brands). Attribute names are correctly singular, (engineSize, email, phone, hourMeterOut).

Group 25, Matthew and Spencer, good job!

Thanks for posting.

This review was done without any AI assistance. All my work.
(moved this from a comment to an answer)

Review by: Devon Carter

Hi Team!

Starting off, I love your overview. It clearly explains an issue, illustrates the scale with precise figures of bikes, brands, customers and rental sessions, and explains how your database will effectively solve the stated problem.

You have the required four entities, each with clearly explained scope and purpose, and matching the figures stated in your overview. Further, your entity details are consistent with what one would expect recorded for this type of business, and your relationships and foreign keys are very clearly explained throughout.

One change I may make would be to replace the varchar in the phone attribute within Customers with int, removing the standard (XXX)XXX-XXXX format, but preventing possible typographic errors which place non-numerical characters in the entry.

Looking through the ERD, it appears consistent with the information presented in your database outline, and neatly organized to be clearly readable. I would suggest removing the M:N connector between Bikes and Customers, as that is expressed through their intersection in Rentals.

1:M relationships are properly expressed through Foreign Keys, and there is an M:N relationship properly expressed using Rentals as an intersection.

Naming of entities is mostly consistent: Entity names are plural and capitalized, and attributes are camel case and mostly singular, with the sole exception of the engineHours attribute under Bikes which should likely be engineHour for consistency.

All work above is original and written by myself.

Review by: Jonathan Marchand

Does the overview describe what problem is to be solved by a website with DB back end? If yes, summarize. If not, what changes would better support describing the problem to be solved?

Yes the overview does describe the problem that is being solved. The problem being Dirty Spokes Dirtbike is in need of a system to correctly manage bike inventory, customer records, and rental history. Using a manual paper record system in no way meets the demands of today's business environment. The implementation of a dbms to correct the problems with data redundancy, data accuracy, as well as the location of valuable inventory definitely seems like a requirement for a rental business where the inventory is the main generator of revenue. I think this is a great idea and you stated your solution very clearly.

Does the overview list specific facts? If yes, summarize what the facts illustrate about the proposed DB solution. If not, what facts would better support illustrating the scope and scale of the proposed DB solution?

Yes the overview lists specific facts specifically the number of customers, rental sessions, bikes in their rental fleet, as well as the number of rental sessions per year. I think what you have so far is sufficient enough to build out a capable dbms for this fleet management issue.

One suggestion would be to incorporate a staff table as well. Data integrity is invaluable and controlling who has access to what features can be an intelligent implementation regarding data security and tracking updates to the dbms. This concept may go beyond this class but the concept of least privilege is very important in cybersecurity/data security.

Are at least four entities described, and does each one represent a single idea to be stored as a list? If yes, summarize. If not, based on the course material, what changes can you suggest to improve?

Yes there are at least 4 entities; Rentals, Brands, Customers, and Bikes.

The many to many relationship featured in their ER diagram is featured by the Rentals table
Many customers can rent many bikes

A Brand can have many Bikes (one to many).

(i.e. bike models per brand, engine sizes),

Each Bike can have many Rentals (one to many).

This is sufficient and meets the requirements for the project. I have no suggestions to improve at this time.

Does the outline of entity details describe the purpose of each, list attribute datatypes and constraints, and describe relationships between entities? If yes, summarize. If not, based on the course material, what changes can you suggest to improve?

Yes the outline of the describes the purpose of each well. It is fairly straightforward the purpose of each table and the relationship with each other.

Some suggestions:

For the Brands table you could change localDealer to just dealer or remove the NOT NULL. Brands go out of business and dealers may not carry the product anymore even if it is in your inventory or for sale on the used market. Keeping the NOT NULL may be a little too strict. Just a suggestion.

For customers:

You could have a row for license and set that to a BOOL, or even insurance and set that to BOOL

Are 1:M relationships correctly formulated? Is there at least one M:M relationship? Does the ERD present a logical view of the database? If yes, summarize. If not, based on the course material, what changes can you suggest to improve?

Yes the 1:M relationships are formulated correctly. Yes there is one M:M relationship (man Customers can rent many bikes and connects through a join table Rentals). The rentals table shows the correct foreign keys (customerID from Customers and bikeID from Bikes).

Is there consistency in a) naming between overview and entity/attributes b) entities plural, attributes singular c) use of capitalization for naming? If yes, summarize. If not, based on the course material, what changes can you suggest to improve?

Yes, the naming convention is very consistent. I don't have any suggestions for improvement at this time.

Review by: TA Razeel Hassan Ibrahim Rasheed

Not sure why did you connect Bikes with customer directly while also stating renatls is intersection table. Intersection table serve to simplify M:N relationships.

Actions Based on Feedback

Accepted Actions

- Add lengths to VARCHAR and INT attributes - VARCHAR lengths, INT signs, and DECIMAL placing have been added to all attributes throughout the Database Outline for greater precision and clarity.
- Make Brand mandatory to Bikes - The Brands→Bikes relationship in the ERD has been updated to reflect a mandatory relationship, as every bike in the fleet must be associated with a brand.
- Remove M:N connector between Customers and Bikes in ERD - The direct line between Customers and Bikes has been removed from the ERD. This relationship is already properly represented through the Rentals intersection table and the direct connector was redundant.
- Add cardinality to connections in the ERD - Proper cardinality markers have been added to all relationship lines in the ERD.
- Edit engineHours to singular - engineHours was not present in the Database Outline so it has been added to the Bikes entity and changed to a singular name, engineHourMeter.

Declined Suggestions

- Add Employees table - The scope of this system is intentionally limited to bike inventory, customer records, and rental history. This is an administrator-facing database tool, not an employee management system. Adding an Employee table would exceed the intended scope of the project.
- phone attribute changed from VARCHAR to INT - We chose to keep phone as VARCHAR since phone numbers may include formatting characters such as dashes, parentheses, spaces, and country code prefixes. Storing phone as INT would risk data loss and overflow on numbers with country codes as well as prevent storing any non-numeric character formatting.
- Make localDealer optional - This suggestion was not employed as recording the nearest outside facility that sells or services a specific brand is a necessity in assembling a fleet of popular models and maintaining them throughout their lifecycle. The scenario of a major brand ceasing operations and no longer being available is highly unlikely, and the discontinuation of a brand would result in its removal from the active fleet entirely rather than keeping it with a null dealer field.
- Add license and insurance attributes to Customers - This suggestion is not a necessity in this context. 50cc MX bikes and increasingly popular electric bikes of comparable sizes are geared toward riders as young as 5 years old who would not have a license or

insurance to record. Beyond this, off-highway vehicles do not require a license or insurance to operate in legal settings such as a designated MX or enduro track. Adding these fields would therefore not be applicable to a significant portion of the customer base and is out of scope for the current system design.

Upgrades to the Draft Version

- Added waiverDate: date, NOT NULL to the Customers entity to record the date a customer signed the facility liability waiver before their first rental.

Overview

Dirty Spokes is a recreational dirtbike rental facility and track located in sunny Southern California, operating a fleet of 50 bikes across 8 major brands including Honda, Yamaha, and Kawasaki. The facility services roughly 500 unique customers per year across an estimated 2,500 annual rental sessions. Currently, staff manage bike inventory, customer records, and rental history through manual bookkeeping and paper waivers.

Over the past year, this has led to a number of operational issues including customer records being lost or duplicated, no reliable way to track when a customer had which bike at any given time, and difficulty identifying when a bike was last rented or how many hours it has accumulated.

A database-driven web application will allow Dirty Spokes administrators to manage Bikes, Brands, Customers, and Rentals in a unified system, ensuring accurate rental records, up-to-date usage metrics for bike maintenance and popularity, and smoother day-to-day operations at the track.

Database Outline

Bikes: records the dirtbikes available in the rental fleet

- bikeID: INT UNSIGNED, AUTO_INCREMENT, unique, NOT NULL, PK
- frameNumber: VARCHAR(50), unique, NOT NULL
- brandID: INT UNSIGNED, NOT NULL, FK → Brands.brandID
- modelName: VARCHAR(100), NOT NULL
- engineSize: VARCHAR(10), NOT NULL - e.g. "250cc", "450cc"
- bikeYear: INT UNSIGNED, NOT NULL
- engineHourMeter: DECIMAL(6,1), NOT NULL
- Relationships:
 - M:1 between Bikes and Brands, many bikes can belong to one brand implemented by FK brandID
 - 1:M between Bikes and Rentals, one bike can appear in many rental records implemented by FK bikeID in Rentals

Brands: records the manufacturers of the bikes in the fleet

- brandID: INT UNSIGNED, AUTO_INCREMENT, unique, NOT NULL, PK
- brandName: VARCHAR(100), unique, NOT NULL
- countryOfOrigin: VARCHAR(100), NOT NULL
- localDealer: VARCHAR(255), NOT NULL
- Relationships:
 - 1:M between Brands and Bikes, one brand can have many bikes implemented by brandID FK in Bikes

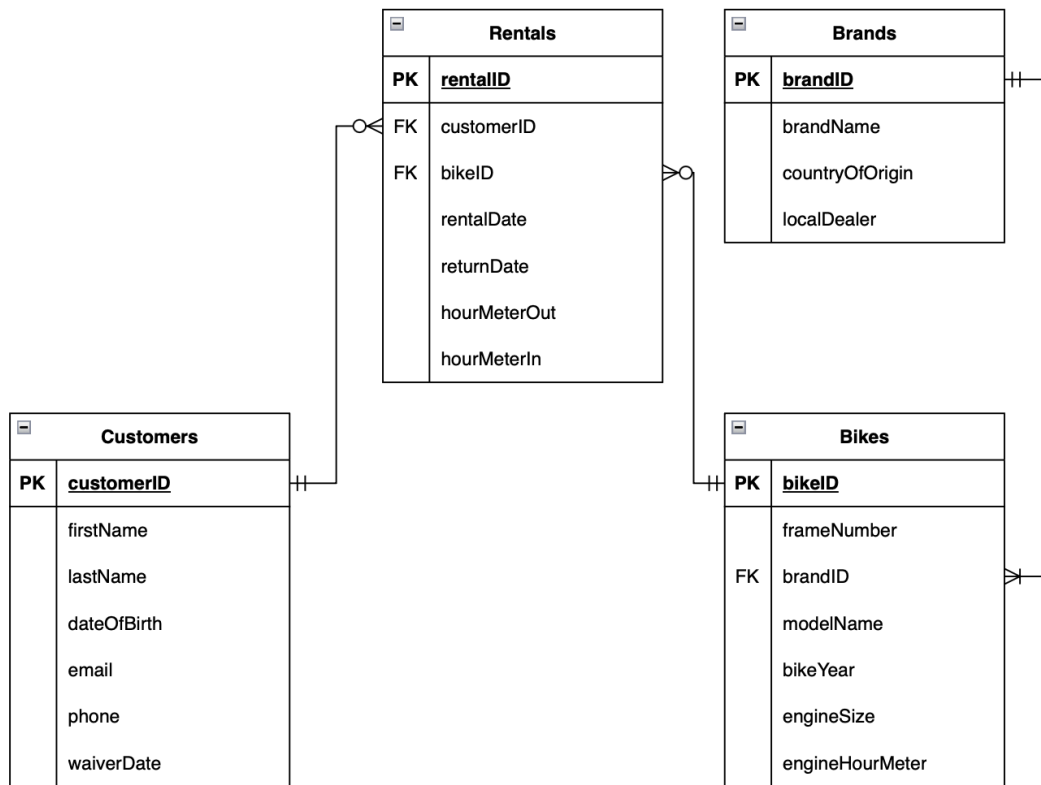
Customers: records customers who rent bikes at the track

- customerID: INT UNSIGNED, AUTO_INCREMENT, unique, NOT NULL, PK
- firstName: VARCHAR(50), NOT NULL
- lastName: VARCHAR(50), NOT NULL
- dateOfBirth: date, NOT NULL
- email: VARCHAR(255), unique, NOT NULL
- phone: VARCHAR(20), NOT NULL
- waiverDate: date, NOT NULL
- Relationships:
 - 1:M between Customers and Rentals, one customer can have many rental records implemented by FK customerID in Rentals

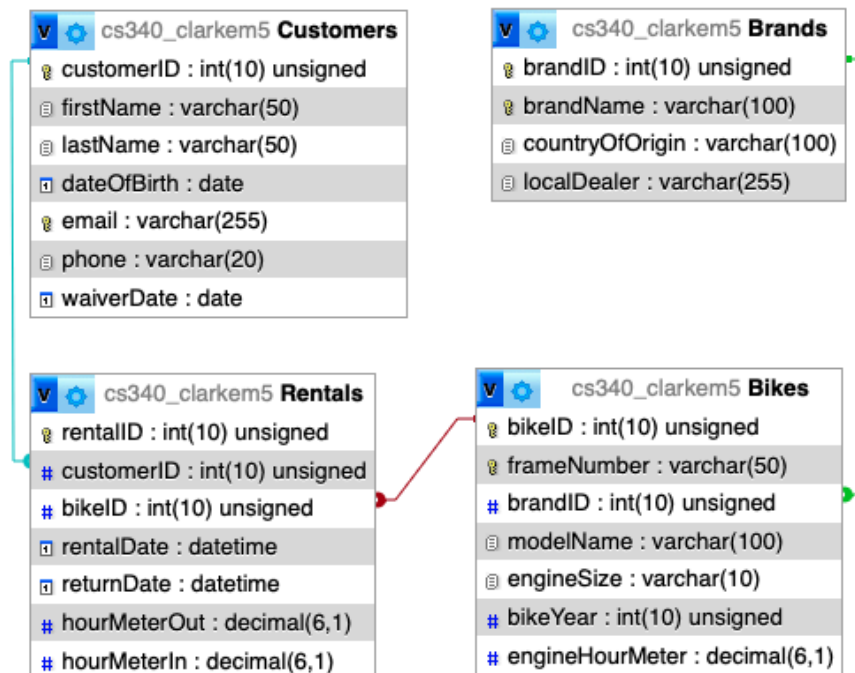
Rentals: intersection table recording which customer rented which bike and when

- rentalID: INT UNSIGNED, AUTO_INCREMENT, unique, NOT NULL, PK
- customerID: INT UNSIGNED, NOT NULL, FK → Customers.customerID
- bikeID: INT UNSIGNED, NOT NULL, FK → Bikes.bikeID
- rentalDate: datetime, NOT NULL
- returnDate: datetime, NOT NULL
- hourMeterOut: DECIMAL(6,1), NOT NULL
- hourMeterIn: DECIMAL(6,1), NOT NULL
- Relationships:
 - M:1 between Rentals and Customers, many rental records can belong to one customer where Customers.customerID is PK and Rentals.customerID is FK
 - M:1 between Rentals and Bikes, many rental records can reference one bike where Bikes.bikeID is PK and Rentals.bikeID is FK
 - M:N between Customers and Bikes, a customer can rent many bikes and a bike can be rented by many customers

ER Diagram



Schema Diagram



Example Data

Bikes Table

bikeID	frameNumber	brandID	modelName	engineSize	bikeYear	engineHourMeter
1	JYACB11C0RA015607	3	YZ65	65cc	2024	15.8
2	VBKXWM236MM321033	2	300 XC-W	293cc	2021	225.5
3	VNBS648C4PB000307	1	SE-F 500 Factory 4T	478cc	2023	71.4

Brands Table

brandID	brandName	countryOfOrigin	localDealer
1	Sherco	France	Trials Offroad
2	KTM	Austria	SoCal Moto
3	Yamaha	Japan	SoCal Moto

Customers Table

customerID	firstName	lastName	dateOfBirth	email	phone	waiverDate
1	Jeremy	Stevens	1989-06-11	jsteve0@email.com	555-0101	2024-11-22
2	Andrea	Hernandez	2003-11-25	andrea44@email.com	555-0202	2025-04-17
3	Roman	Lutoslawski	1978-08-26	roman444@email.com	555-0303	2020-08-20

Rentals Table

rentalID	customerID	bikeID	rentalDate	returnDate	hourMeterOut	hourMeterIn
1	3	3	2025-04-06 11:22:56	2025-04-08 16:43:16	219.9	225.5
2	3	2	2026-04-10 09:30:17	2026-04-13 10:17:44	60.6	68.0
3	2	2	2026-04-19 10:19:19	2026-04-19 15:22:04	68.1	71.4

Citations

[1] *Motocross Wallpaper* (April 6, 2026) wallpapers.com <https://wallpapers.com/motocross>

All other work in this proposal was done without the use of AI tools.